Total No. of Printed Pages: 2

SUBJECT CODE NO: - RNP-8064
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (EEE / EE / EEP) (NEP) Sem - IV
Examination May/June 2026
Network Analysis

[Time: 1:00 Hour] **[Max. Marks: 30]**

Please check whether you have got the right question paper.

N. B
1) Part A: Question No 1 is compulsory.
2) Attempt any four questions from remaining Part B: Q.2 to Q.7.

Part A

Q.1 Solve any 5 following questions. **10**

1) Z-parameters are also called:
A) Impedance Parameters B) Admittance Parameters
C) Hybrid Parameters D) Transmission Parameters

2) A tree in graph theory contains:
A) All loops B) No loops
C) Only nodes D) Only branches

3) At resonance in a series RLC circuit:
A) XL > XC B) XC > XL
C) XL = XC D) XL = 0

4) Node analysis is based on:
A) KVL B) KCL
C) Ohm's Law D) Superposition

5) Thevenin's theorem replaces a network by:
A) Current Source
B) Voltage Source in Series with Resistance
C) Only Resistance
D) Only Voltage

6) Maximum power transfer occurs when:
A) RL = 0 B) RL = Rth
C) RL = 2Rth D) RL = Rth/2

7) Y- Parameters are also known as:
A) Impedance Parameter B) Admittance Parameter
C) Reactance Parameter D) None of the above

8) Time constant of an RC circuit is:
A) R/C B) RC C) 1/RC D) R^2C

Part B

Solve any four from the following

Q.2 Explain Thevenin's theorem with procedure. **05**

Q.3 State and explain superposition theorem. **05**

Q.4 Define graph, tree, and co-tree in network theory. **05**

Q.5 Explain symmetrical and asymmetrical networks. **05**

Q.6 Compare series resonance and parallel resonance (impedance, current, power factor. **05**

Q.7 Explain two-port network Z parameters. **05**